INCH-POUND
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SUPERSEDING
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MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, M99 REMANUFACTURED METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for M99 Cartridge Actuated Initiator Remanufactured Metal Parts Assembly *(see 6.1)*.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1377

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (CAGE Code 19200 and 19203)

10520406	Initiator, Cartridge Actuated, M99, Metal Parts Assembly
8830984	Box, Fiberboard (Inner) Packing, Ammunition for Initiator, Cartridge Actuated, M27, M86, M87, M98, M99, or XM107
8830985	Box, Packing, Ammunition for Initiator, Cartridge Actuated, M27, M86, M87, M98, M99, or XM107
9241155	Box, Fiberboard (Outer) Alternative Packing Ammunition for Initiator, Cartridge Actuated, M27, M86, M87, M98, M99, or XM107

U.S. AIR FORCE

TPO 00-845-1059	Initiator, Cartridge Actuated, M99

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian Head, MD 20640-5000.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefore has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. The remanufactured initiator metal parts assembly shall contain a new seal, initiator spring, O-ring, and safety pin.

3.2 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component, regardless of configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied (see 6.3.2).

3.3 Hydrostatic pressure. The initiator chamber and cap shall withstand a hydrostatic pressure of 14,000 to 15,000 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.3.1).

3.4 Firing pin release. The firing pin of the metal parts assembly, less safety pin, shall release from cocked position when the application of an axial load of 18 to 25 pounds has withdrawn the initiator pin 0.370 inches, minimum, and 0.425 inches, maximum.

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3.5 Initiator pin security. Application of axial load of 200 to 220 pounds to the initiator pin of the metal parts assembly shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin.

3.6 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.6.1 Integrated plant products. When remanufacturing and loading of assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

3.6.2 Threaded component parts. Class 2A or 2B thread gage shall be used on component part inspection versus class 3A or 3B required on remanufactured hardware for initiators.

3.7 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than 5 degrees in either direction using procedures in 4.5.5.

3.8 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

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4.1.3 Inspection. Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 20 to 25 °C (68 to 77 °F)
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum.
d. Atmospheric pressure: 14.7 psia

4.3 Quality conformance inspection.

4.3.1 Lot.

4.3.1.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 initiators, maximum, provided the homogeneity requirements of inspection lots have been retained.

4.3.2 Examination. One-hundred percent examination of reprocessed parts shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.3.2.1 Examination of reclaimed parts. Examination of reuseable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.3.2.2 Chamber, Initiator, Drawing 10520407, covering a detail of Drawing 10520406.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Pitch diameter of large external thread, min	Gage
102	Major diameter of large external thread, min	Gage
103	Pitch diameter of small external thread, min	Gage
104	Major diameter of small external thread, min	Gage
105	Pitch diameter of internal thread, max	Gage
106	Minor diameter of internal thread, max	Gage
107	Thread damaged	Visual
108	Metal defective	Visual
109	Protective finish, inadequate or defective	Visual
110	Surface finish of cavity and angle of gas port, improper	Visual
111	Burr	Visual
112	Foreign matter present	Visual

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Characteristic	Defect	Method of Inspection
Minor		
201	Surface finish, improper, except as otherwise classified	Visual

4.3.2.3 Cap, Initiator, Drawing 10520408, covering a detail of Drawing 10520406.

Critical		
None defined		
Major		
101	Pitch diameter of external thread, min	Gage
102	Major diameter of external thread, min	Gage
103	Effective thread length of external thread, min	Gage
104	Pitch diameter of small internal thread, max	Gage
105	Minor diameater of small internal thread, max	Gage
106	Pitch diameter of large internal thread, max	Gage
107	Minor diameter of large internal thread, max	Gage
108	Metal defective	Visual
109	Protective finish, inadequate or defective	Visual
110	Burr	Visual
111	Foreign matter present	Visual
Minor		
201	Surface finish improper	Visual

4.3.2.4 Pin, Initiator, Drawing 10520409, covering a detail of Drawing 10520406.

Critical		
None defined		
Major		
101	Metal defective	Visual
102	Protective finish, inadequate or defective	Visual
103	Surface finish of groove in head and face of shoulder, improper	Visual
104	Burr	Visual
105	Foreign matter present	Visual
Minor		
201	Surface finish, improper, except as otherwise classified	Visual

4.3.2.5 Housing, Firing Pin, Drawing 10520410, covering a detail of Drawing 10520406.

Critical		
None defined		

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Characteristic	Defect	Method of Inspection
Major		
101	Thread damaged	Visual
102	Metal defective	Visual
103	Protective finish, inadequate or defective	Visual
104	Surface finish of cavity, improper	Visual
105	Burr	Visual
106	Foreign matter present	Visual
Minor		
201	Pitch diameter of thread, min	Visual
202	Major diameter of thread, min	Visual

4.3.2.6 Pin, Firing, Drawing 10520411, covering a detail of Drawing 10520406.

Critical		
None defined		
Major		
101	Metal defective	Visual
102	Protective finish, inadequate or defective	Visual
103	Surface finish of body, improper	Visual
104	Burr	Visual
105	Foreign matter present	Visual
Minor		
None defined		

4.3.2.7 Filter, Drawing 10520414, covering a detail of Drawing 10520406.

Critical		
None defined		
Major		
101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Thread damaged	Visual
104	Nylon patch not properly located or fused	Manual/Visual
105	Metal defective	Visual
106	Protective finish, inadequate or defective	Visual
107	Burr	Visual
108	Foreign matter present	Visual
Minor		
None defined		

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4.3.2.8 Assembly, Drawing 10520406, less all Drawings 10520410, 10520411, and MS 19060-3.

Characteristic	Defect	Method of Inspection
Critical		
1	Firing pin protrusion	Gage
2	Steal ball missing	Gage
3	O-ring missing	Visual
4	Seal missing	Visual
Major		
101	Burr	Visual
102	Foreign matter present	Visual
Minor		
None defined		

4.3.2.9 Assembly, Drawing 10520406.

Critical		
None defined		
Major		
101	Firing pin not cocked	Manual
102	Initiator pin not free to rotate	Manual
103	Any part missing, except as otherwise classified	Visual
104	Improper assembly	Visual
105	O-ring not lubricated	Visual
Minor		
201	Protective cap missing	Visual
202	Protective finish, inadequate or defective	Visual

4.3.2.10 Box, Fiberboard (Inner) Packing Ammunition, Drawing 8830984.

Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Sealing strip missing or incorrectly applied	Visual
204	Marking missing, incorrect or illegible	Visual
205	Cut, damaged or improperly sealed bag	Visual

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4.3.2.11 Box, Packing Ammunition, Drawing 8830985.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Strapping missing, loose, broken or improperly assembled	Manual/Visual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect or illegible	Visual
206	Number of units incorrect	Visual

4.3.2.12 Box, Fiberboard (Outer) Alternative Packing Ammunition, Drawing 9241155.

Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.3.2.13 Seal, Drawing 10520415 covering a detail of Drawing 10520406

Critical		
None defined		
Major		
101	Diameter, outside	Gage
102	Thickness	Gage
103	Hardness	Gage
104	Flash	Visual
105	Cut, cracked, or split	Visual
106	Foreign matter	Visual
Minor		
201	Wall thickness	Gage

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4.3.2.14 Spring, Initiator, Drawing 10520412 covering a detail of Drawing 10520406.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Load at long compressed length	Gage
102	Load at short compressed length	Gage
103	Solid length, max	Gage
104	Diameter, outside	Gage
105	Ends incorrect	Visual
106	Protective finish inadequate or defective	Visual
107	Burr	Visual
108	Foreign matter present	Visual
Minor		
None defined		

4.3.2.15 Pin, Safety, Drawing 10522817 covering a detail of Drawing 10520406.

Critical		
None defined		
Major		
101	Diameter of wire	Gage
102	Profile incorrect	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter present	Visual
Minor		
201	Diameter of head	Gage
202	Total length	Gage

4.4. Inspection testing. The following tests shall be performed in accordance with the provisions of 4.5.

4.4.1 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.4.2 Hydrostatic pressure. Each chamber and cap subassembly shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with the requirement of 3.3 shall be cause for rejection of the subassembly.

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4.4.3 Firing pin release. The initiator pin of each partial assembly consisting of all parts except the safety pin and chamber shall be subjected to the firing pin release test. Failure of any partial assembly to comply with the requirement of 3.4 shall be cause for rejection of the subassembly.

4.4.4 Initiator pin security. The initiator pin of each subassembly consisting of all parts except the chamber, shall be subjected to the initiator pin security test. Failure of any subassembly to comply with the requirements of 3.5 shall be cause for rejection of the subassembly.

4.4.5 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit in the inspection sample to comply with the requirement of 3.7 shall be cause for rejection of the assembly.

4.5 Test methods and procedures.

4.5.1 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.5.2 Hydrostatic pressure. Prior to the application of protective finish, the cap and chamber subassembly shall be mounted in the device, using a fixture for properly sealing all openings and subjected to the internal hydrostatic pressure to determine compliance with the requirement of 3.3.

4.5.3 Firing pin release. The initiator pin of each partial assembly consisting of all parts except the safety pin and chamber shall be subjected to the firing pin release test. Failure of any partial assembly to comply with the requirements of 3.4 shall be cause for rejection of the subassembly. To achieve the pull force requirements of 18-25 pounds specified in paragraph 3.4 selective reassembly may be necessary (i.e., changing initiator pins) for those assemblies which are initially out of range.

4.5.4 Initiator pin security. The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.5.5 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for 4 full passes.

4.6 Inspection of packaging. The sampling and inspection of the packing and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING.

5.1 Packing and marking. Packing and marking shall be levels A or B as specified by the contracting activity:

Level A	In accordance with Drawings 8830984 and 8830985
Level B	In accordance with Drawings 8830984 and 9241155

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-845-1059.

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6. Notes.

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

6.2 Acquistion requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
c. The responsible inspection activity (see 4.1)
d. Any special inspection conditions (see 4.3)
e. Level of marking and packing (see 5.1).

6.3 Definitions.

6.3.1 Mechanical failure. A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3.2 Remanufactured exterior components. Remanufactured exterior components are: initiator pin, cap, and chamber.

6.4 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape system, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy-OS
Air Force-99

Review activity
Air Force-70

Preparing activity:
Navy-OS
(Project 1377-N945)

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